

Chapter 5. Factor groups

5.3 The First Isomorphism Theorem

$$\begin{array}{ccc} G & \xrightarrow{f} & H \\ \pi \downarrow & \nearrow \tilde{f} & \\ G / \ker(f) & & \end{array}$$

In this video, we will prove a very important result known as the First Isomorphism Theorem.

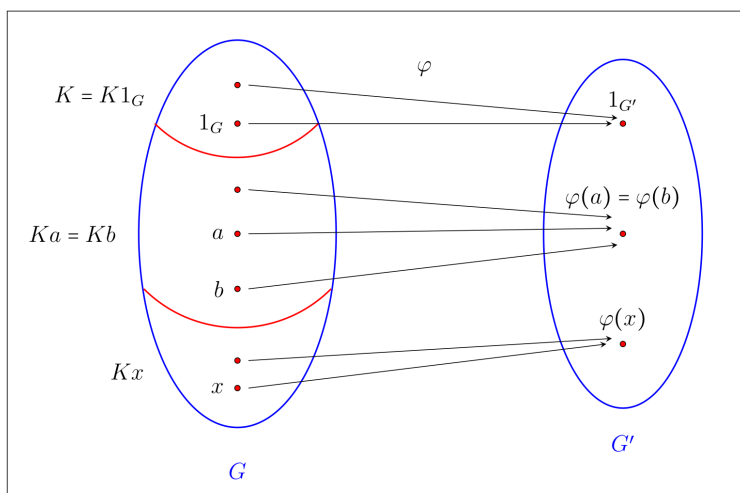
Proving the First Isomorphism Theorem

Let G and G' be two groups. We say that G' is a **homomorphic image** of G if there exists an epimorphism from G onto G' . Using this terminology, the canonical projection (or coset mapping) tells us that **any factor group G/H is a homomorphic image of G** .

$$\pi : G \rightarrow G/H \quad \pi(a) = Ha$$

In other words, the factor group construction gives us a way of manufacturing homomorphic images of a group G . In this section, we will show that this is indeed a way of manufacturing all the homomorphic images of G . To be more precise, we will prove that **every homomorphic image of G is isomorphic to a factor group of G** .

THEOREM 5.11. *Let $\varphi : G \rightarrow G'$ be a group homomorphism with kernel K . Then $\varphi(a) = \varphi(b)$ if and only if $Ka = Kb$.*



Proof. For a and b in G , we have that

$$\varphi(a) = \varphi(b) \Leftrightarrow \underline{1_{G'}} = \varphi(a) \underbrace{\varphi(b)^{-1}}_{\parallel \varphi(b^{-1})} = \varphi(a) \varphi(b^{-1})$$

$$\begin{array}{l} \text{def. of kernel} \\ \downarrow \\ \Leftrightarrow ab^{-1} \in K \end{array} \quad = \underline{\varphi(ab^{-1})}$$

$$\begin{array}{l} \uparrow \\ \text{coset} \\ \text{properties} \end{array} \quad \Leftrightarrow Ka = Kb$$

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THEOREM 5.12. FIRST ISOMORPHISM THEOREM. Let $\varphi: G \rightarrow G'$ be a group epimorphism with kernel $K = \ker \varphi$. Then $G/K \cong G'$.

Proof. We need to define a group homomorphism that is bijective from G/K onto G' .

$$\tilde{\varphi}: G/K \longrightarrow G'$$

$$\tilde{\varphi}(Kg) = \varphi(g) \text{ for all } g \in G.$$

If $Kg = Kg'$, then by Thm 5.11 we have that $\varphi(g) = \varphi(g')$. So this shows that $\tilde{\varphi}$ does not depend on the generator of the coset.

•) $\tilde{\varphi}$ is injective:

$$\tilde{\varphi}(Ka) = \tilde{\varphi}(Kb) \Rightarrow \varphi(a) = \varphi(b)$$

Thm 5.11

$$\downarrow \\ \Rightarrow Ka = Kb$$

•) $\tilde{\varphi}$ is surjective:

For any $g' \in G$, we can find $a \in G$ such that (φ is surjective)

$$g' = \varphi(a) \stackrel{\text{def}}{=} \tilde{\varphi}(Ka)$$

•) $\tilde{\varphi}$ is a group homomorphism:

$$\tilde{\varphi}((Ka)(Kb)) \stackrel{?}{=} \tilde{\varphi}(Ka) \cdot \tilde{\varphi}(Kb)$$

$$\tilde{\varphi}((Ka) \cdot (Kb)) \stackrel{\text{coset op.}}{=} \tilde{\varphi}(K(ab)) \stackrel{\text{def } \tilde{\varphi}}{=} \varphi(ab)$$

$$\stackrel{\uparrow}{=} \varphi(a)\varphi(b) \stackrel{\text{def } \tilde{\varphi}}{=} \tilde{\varphi}(Ka)\tilde{\varphi}(Kb)$$

φ is a group homom.



The *First Isomorphism Theorem* asserts that **every homomorphic image of G is isomorphic to a quotient group of G** . If we are given an homomorphism $\varphi : G \rightarrow G'$, not necessarily surjective, we can consider the restriction $\varphi : G \rightarrow \varphi(G)$ to the image, which is now an epimorphism, and apply the *First Isomorphism Theorem* to it. More precisely:

COROLLARY 5.13. *Let $\varphi : G \rightarrow G'$ be a group homomorphism with kernel K . Then $G/K \cong \text{Im}\varphi$.*